Here are some explanations for some questions in problem sheet 2.

Q 1. A group is said to be a finite group if it has only finitely many elements. You can take the product of elements in any order (in the question 1).

Q 4. The law of composition on $\mathbb{Z}\times\mathbb{Z}$ is defined as $(a,b)(c,d) = (a+c, b+d)$ for all $a, b, c, d \in \mathbb{Z}$.

Q 13. New product p_m in M is the new law of composition on M. Unit relative to p_m means that it is a neutral/identity element in the associative magma (M, p_m).

(In the previous mail, I replied that a unit element in a group/monoid means that it is an invertible element, which is incorrect. I am sorry about that.)

Q 7. In the question 7, prove or disprove the statement.

On the product set $\mathbb{R} \times \mathbb{R}$, the law of composition $*$ is componentwise addition:

$$(a, b) * (c, d) = (a + c, b + d) \quad \text{for all} \quad a, b, c, d \in \mathbb{R}.$$

On the product set, $\mathbb{R}^+ \times \mathbb{C}$ the law of composition $*$ is defined as

$$(r, u) * (s, v) = (r.s, u + v) \quad \text{for all} \quad r, s \in \mathbb{R}^+ \text{and} \, u, v \in \mathbb{C}.$$

(**Note :** Here $\mathbb{R}^+$ denote the group of positive real numbers with multiplication ($\mathbb{R}^+, \cdot$))

Note: Let $(G, *_1)$ and $(H, *_2)$ be two groups. Consider the law of composition $*$ on the set $G \times H$ defined as $(g_1, h_1) * (g_2, h_2) = (g_1 *_1 g_2,\ h_1 *_2 h_2)$ for all $g_1, g_2 \in G$ and $h_1, h_2 \in H$ where $*_1$ and $*_2$ are the laws of composition on G and H respectively. Then $(G \times H,\ *)$ is a group (verify as an exercise), and it is called the product group of G and H.